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NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)
Fourth Semester B.Tech (CBCS) Degree Examinations

May 2025

CC2102-1 – FOUNDATIONS OF OPERATING SYSTEMS

(For CC)

Duration: 3 Hours

Max. Marks: 100

Note:

Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

Part – B: Descriptive Answer Questions: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II each** and **One full** question from **Unit – III**.

PART - A: MULTIPLE CHOICE QUESTIONS

20 Marks

1. **What is the main purpose of a linker?**
A) To convert assembly code to machine code B) To load a program into memory for execution
C) To resolve symbol references and combine object files D) To interpret high-level language instructions
2. **The main advantage of a microkernel architecture is:**
A) High efficiency and fast performance B) Reduced size and better security
C) Direct communication with hardware D) No need for user programs
3. **The dispatcher is responsible for:**
A) Scheduling I/O-bound processes B) Swapping processes between disk and main memory
C) Switching context and transferring control to the selected process D) Allocating disk space
4. **The time taken for a process to execute from start to finish is called:**
A) Waiting time B) Turnaround time
C) Response time D) Dispatch latency
5. **A critical section in a process is a part where:**
A) The process updates shared resources B) The process executes privileged instructions
C) The process performs only I/O operations D) The process enters an infinite loop
6. **Deadlock avoidance requires:**
A) Knowledge of resource allocation in advance B) Preempting resources from processes
C) Assigning fixed priorities to processes D) Ignoring circular wait conditions
7. **A page table is used for:**
A) Storing memory addresses of running processes B) Mapping logical addresses to physical addresses
C) Storing the list of all processes in memory D) Tracking CPU scheduling
8. **Thrashing occurs when:**
A) A process spends more time paging than executing B) The system has too much free memory
C) The page fault rate is very low D) More frames are allocated to a process

9. **Page replacement algorithms are required when:**
 A) There is no free frame available in memory B) Processes are running in kernel mode
 C) Swapping is disabled D) CPU scheduling is inefficient
10. **Which of the following is true about swapping?**
 A) It reduces memory fragmentation B) It moves entire processes between disk and memory
 C) It increases execution speed D) It replaces page tables with segment tables
11. **Which of the following best describes the function of copy-on-write?**
 A) A process immediately writes changes to memory B) Processes share pages until a write occurs, at which point a copy is made
 C) The system uses only physical memory for execution D) It is a technique for swapping pages
12. **Which of the following is NOT a type of system call?**
 A) Process control B) Device management
 C) Machine learning D) File management
13. **The process scheduler is responsible for**
 A) Managing CPU registers B) Allocating CPU time to processes
 C) Performing I/O operations D) Handling memory management
14. **What is the function of the fork() system call in Unix-based OS?**
 A) Create a new process B) Terminate a process
 C) Switch between processes D) Allocate memory
15. **Which of the following is NOT an access method for files?**
 A) Sequential access B) Direct access
 C) Random access D) Indirect access
16. **In an acyclic-graph directory structure, a file can:**
 A) Have multiple parent directories B) Be accessed only sequentially
 C) Be linked to by only one directory D) Have no symbolic links
17. **The access control list (ACL) is used to:**
 A) Allocate memory for files B) Manage file ownership
 C) Define which users can access a file D) Store file attributes
18. **Which directory implementation technique improves search efficiency?**
 A) Linear list B) Linked list
 C) Hash table D) Stack
19. **Which of the following optimizes virtualization performance?**
 A) Binary translation B) Direct execution
 C) Memory swapping D) Disk fragmentation
20. **What is the purpose of I/O virtualization?**
 A) Improve input/output device sharing among virtual machines B) Reduce CPU load
 C) Increase disk fragmentation D) Prevent I/O buffer overflow

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit – I		Marks	BT*	CO*	PO*
1.	a) Explain the system calls with examples. Explain the categories of system calls used in an operating system.	8	L*2	1	1,3
	b) Discuss different operating system structures, including monolithic systems, layered architecture, microkernel systems.	8	L2	3	1,3
2.	a) What is process scheduling? Explain different levels of scheduling and their significance.	8	L3	2	1,3

- b) Given the following process set and time quantum = 4 ms:

Process	Arrival Time	Burst Time
P1	0 ms	10 ms
P2	2 ms	4 ms
P3	3 ms	5 ms
P4	5 ms	6 ms

Perform Round Robin scheduling and FCFS scheduling and determine:

i. Gantt Chart

ii. Turnaround Time (TAT)

iii. Waiting Time (WT)

iv. Average Turnaround and Waiting Times

8 L2 3 1,3

3. a) Explain the message passing and shared memory mechanisms. Compare their advantages and disadvantages in IPC.

8 L2 1 1,3

- b) Consider the following processes with given priority values (lower number = higher priority):

Process	Arrival Time	Burst Time	Priority
P1	0 ms	6 ms	3
P2	1 ms	8 ms	1
P3	2 ms	7 ms	2
P4	3 ms	3 ms	4

Schedule using Priority Scheduling (Non-Preemptive) and calculate:

i. Completion Time (CT)

ii. Turnaround Time (TAT)

iii. Waiting Time (WT)

iv. Average Turnaround and Waiting Time

8 L3 2 1,3

Unit – II

4. a) Define deadlock. Discuss necessary and sufficient conditions for a deadlock to occur.
- b) Consider a system with 4 processes (P0 – P3) and 3 resource types (A, B, C). The available resources are: Available=(2,3,2)

8 L2 3 1,3

The maximum demand and allocated resources are:

Process	Max (A, B, C)	Allocation (A, B, C)
P0	(6, 4, 3)	(1, 2, 2)
P1	(3, 2, 2)	(1, 0, 0)
P2	(9, 5, 4)	(3, 3, 2)
P3	(7, 4, 3)	(2, 1, 1)

i) Calculate the Need matrix.

ii) Determine whether the system is in a safe state using the Banker's Algorithm. If yes, find the safe sequence.

8 L3 4 1,3

5. a) Illustrate the structure of a page table and discuss different types of page tables such as hierarchical paging, hashed page table, and inverted page table.
- b) Describe page replacement algorithms (FIFO, LRU, Optimal). Compare their performance with an example.

8 L2 4 1,3

8 L2 4 1,3

6. a) Describe deadlock detection in a system with multiple instances of resources. How is recovery handled once a deadlock is detected?
- b) Explain First Fit, Best Fit, and Worst Fit allocation strategies. A system uses contiguous memory allocation and has a total memory of 100 MB. The memory is allocated as follows:

Process	Size (MB)
P1	20
P2	30
P3	10

A new process P4 (25 MB) arrives.

- i) Can First Fit, Best Fit, and Worst Fit allocation strategies allocate this process?
- ii) If not, how much external fragmentation is present?

8 L2 4 1,3

8 L3 4 1,3

Unit – III

7. a) Explain sequential access, direct access, and indexed access methods with examples.
- b) Describe different file protection mechanisms used in operating systems.
8. a) Compare Contiguous, Linked, and Indexed file allocation methods with diagrams.
- b) Discuss I/O Virtualization techniques used in modern cloud computing.

10 L2 5 1,3

6 L2 5 1,3

8 L2 5 1,3

8 L2 5 1,3

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome.
